

Continual Causal Abstraction

- Why do we need causal abstractions?

Example: imagine being a chemist analyzing a particular gas. Your task $\rightarrow$ analyze temperature & pressure in the volume. You use thermo/baro meter & get answer. Next, you are asked to measure the average velocity of a particle in the gas.

- Previous model becomes obsolete, you now need microscopic level of detail $\rightarrow$ Need different level of abstraction
- New model still needs to be consistent with the previous one (If we measure net kinetic energy of all relevant particle combinations $\rightarrow$ we would ideally like to see a match with the previous measurements of the thermometer)
- Dietitian analyzing causal effect of a particular diet onto the risk of heart disease
Total cholesterol $\longrightarrow$ Heart disease
• balanced diet $\rightarrow$ lower cholesterol $\rightarrow$ Lower risk of HD
- structural equations: dictate how each endogenous variable is caused through a given set of other variables in

terms of an equation.

Example: $HD = f_i(TC, U_i)$

• Here f_i is the mathematical description of the mechanism that makes TC affect HD.

- If for 2 SCMs (M_L, M_H) we have following conditions fulfilled for functions

$$\tau: R_L(V_L) \rightarrow R_H(V_H), \tau_u: R(U_L) \rightarrow R(U_H)$$

$$\& w_T: I_L \rightarrow I_H$$

① All functions are surjective, w_T order preserving

$$\textcircled{2} \forall i \in I_L \text{ we have } \tau(p_{r_L}^i) = p_{r_H}^{w_T(i)}$$

$$\textcircled{3} \tau(M_L(u_L, i)) = M_H(\tau_u(u_L), w_T(i))$$

for all $(u_L, i) \in U_L \times I_L$.